19: Lookup Table Approximation

Engr 315: Hardware / Software Codesign

Andrew Lukefahr
Indiana University

Some material taken from EECS370 at U. of Michigan

The Zoom / Canvas Lecture Plan

- Lecture using Zoom:
  - https://iu.zoom.us/j/362983973
  - Password: ENGR315

- Also available on Canvas
  - Next day -ish

Office Hours

- Ethan is hosting virtual office hours:

- Wed 5:30 PM - 8:00 PM, Th 2:30 PM - 4:00 PM
- https://iu.zoom.us/j/770949556

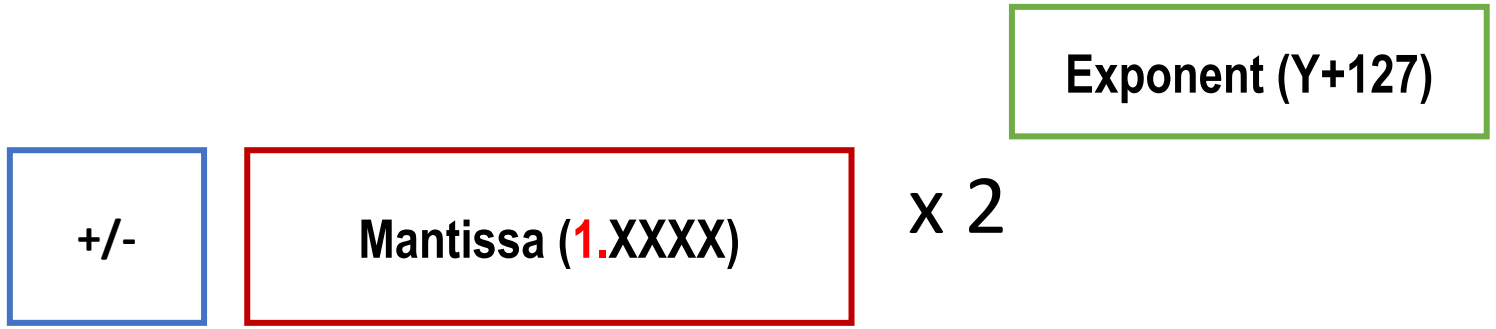
P4: Bitcounting

- Due on Friday
- Austin is going guest lecture
  - Present C implementation.
  - Build a simple Linux kernel module to help

PX: Neural Networks

- We're going to scale this back a little.
- Focus on Tanh ~~Sigmoid~~
- Beyond that is going to be extra credit.

Review: IEEE-754 “Float” Standard



Float Example

Exponent (Y+127)

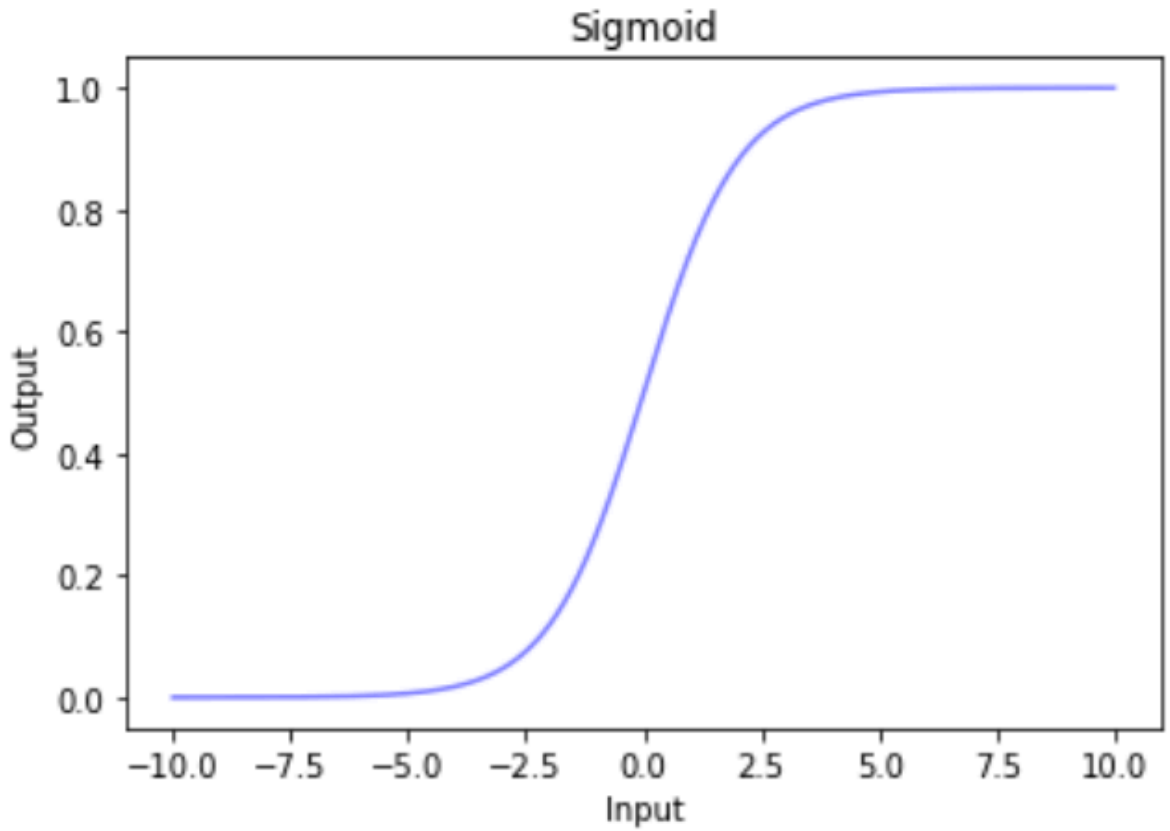
+/- | Mantissa (1.XXXX) | x 2

- $+10.625_{10}$ => $+1010.101_2$ => $+1.010101 \times 2^3$

- Sign: + => 0_2
- Mantissa: ~~1~~.**010101** => **0101 0100 0000 0000 0000 000$_2$**
- Exponent: $3_{10} + 127_{10} = 130_{10}$ => **1000 0010$_2$**

10.625_{10} = 0 10000010 01010100000000000000000

Sigmoid is Floating-Point




```
def sigmoid(x):
    return 1 / (1 + np.e ** -x)
```

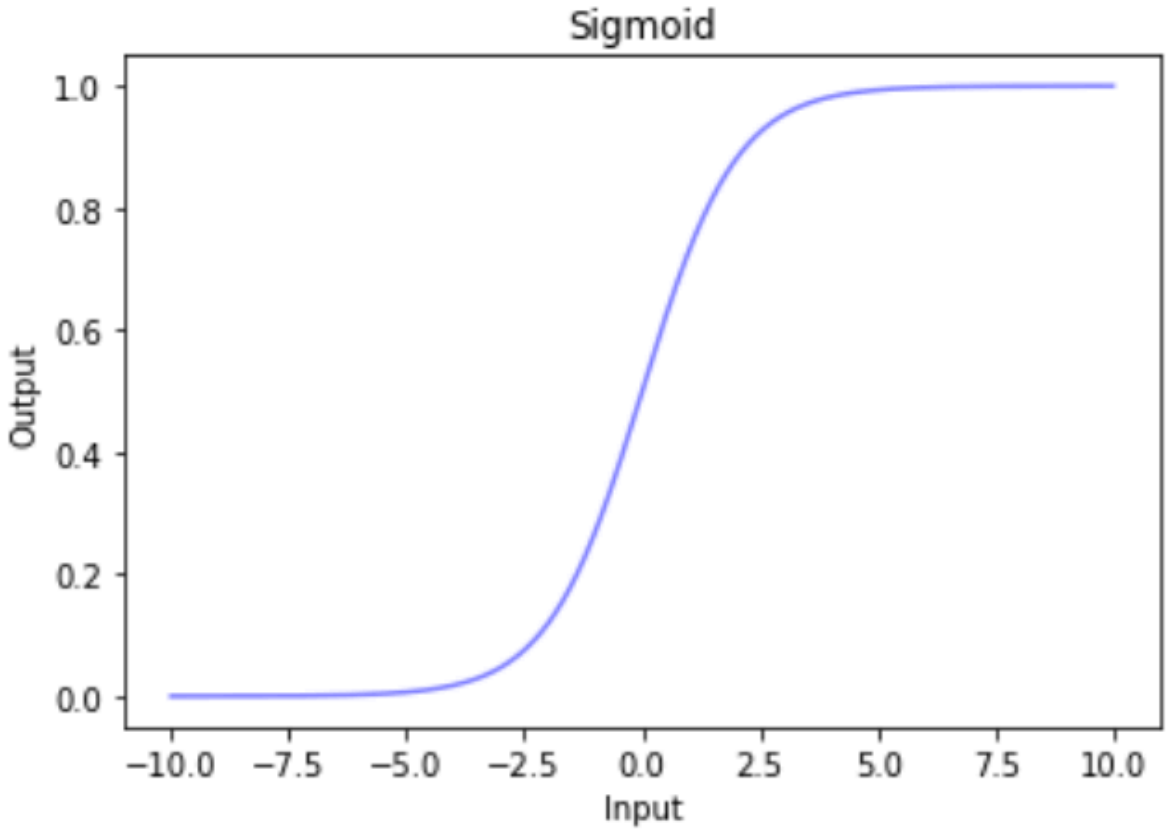
Sigmoid Computation

```
def sigmoid(x):
    return 1 / (1 + np.e ** -x)
```

- 1 floating-point exponential
- 1 floating-point addition
- 1 floating-point division

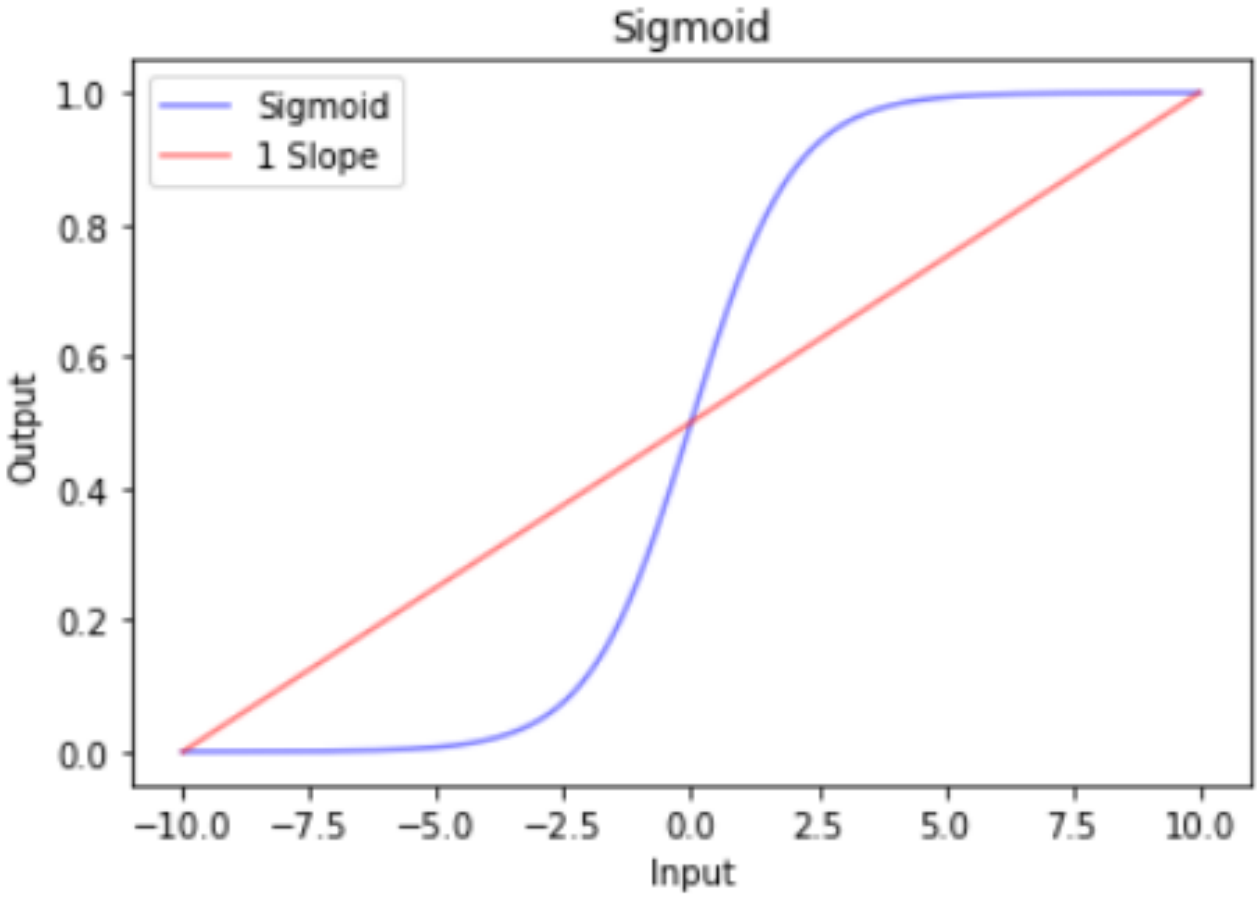
- Can we make this simpler?

Approximating Sigmoid

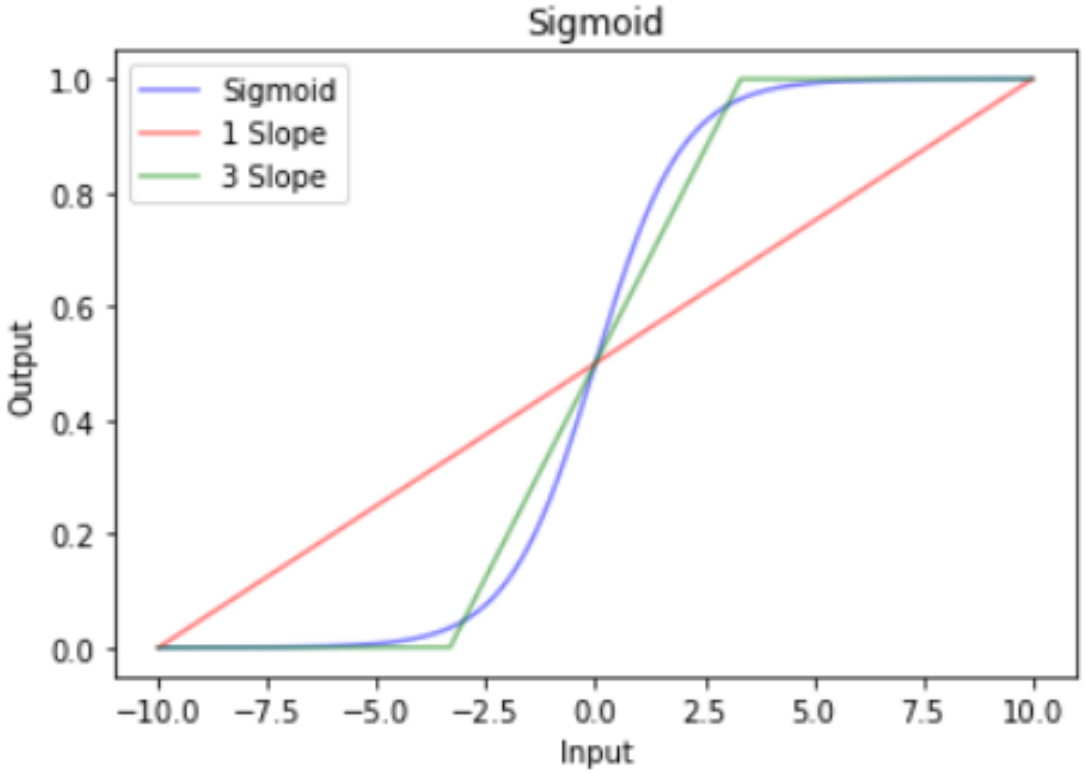


Ways to approximate Sigmoid?

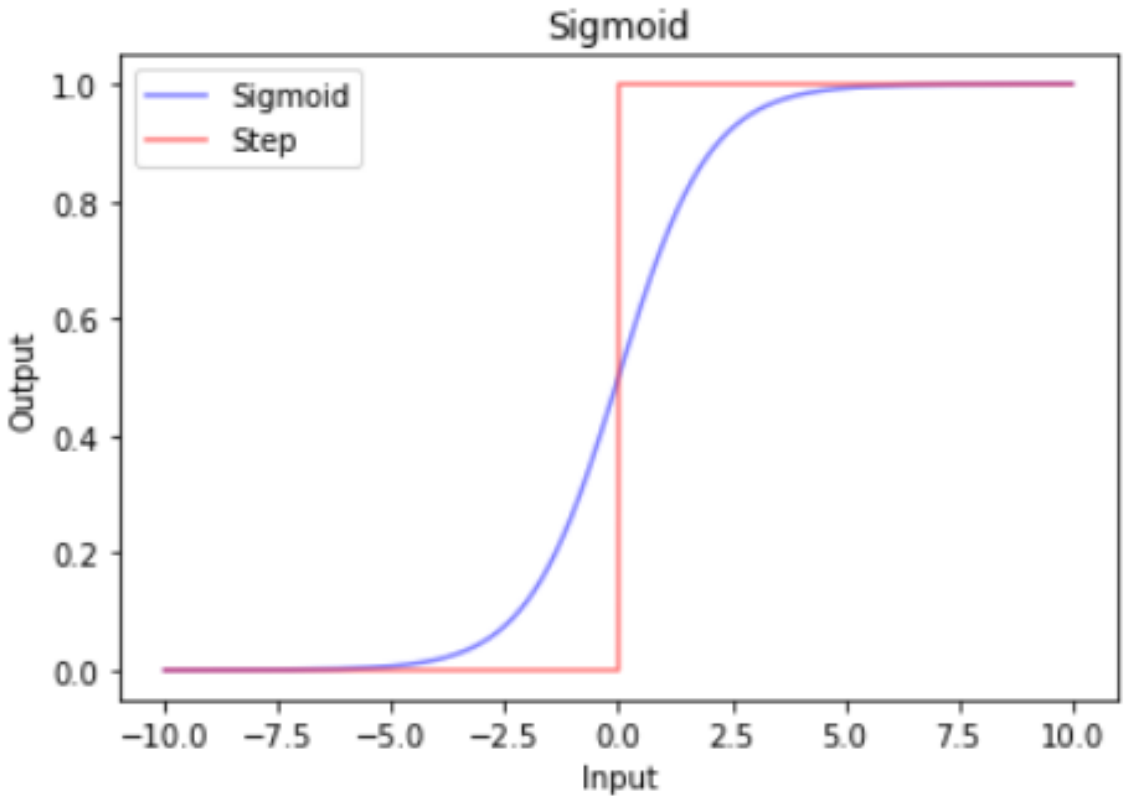
Linear Approximation



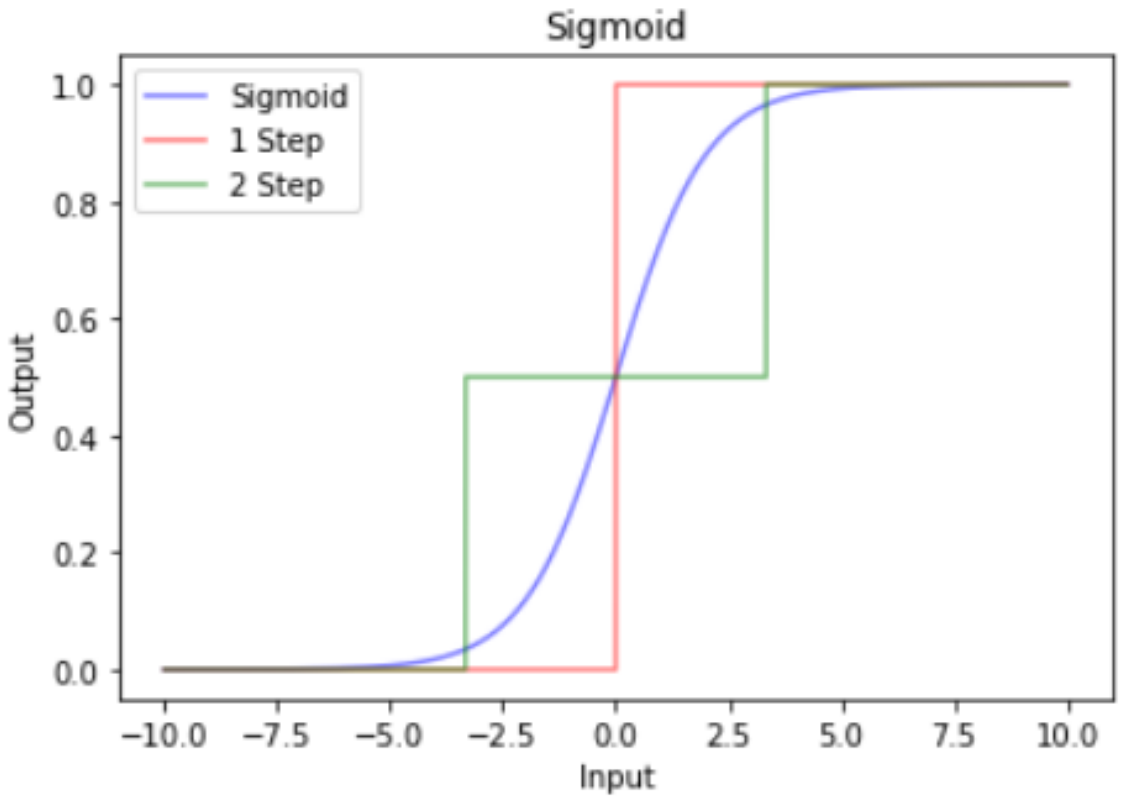
Piece-wise Linear Approximation



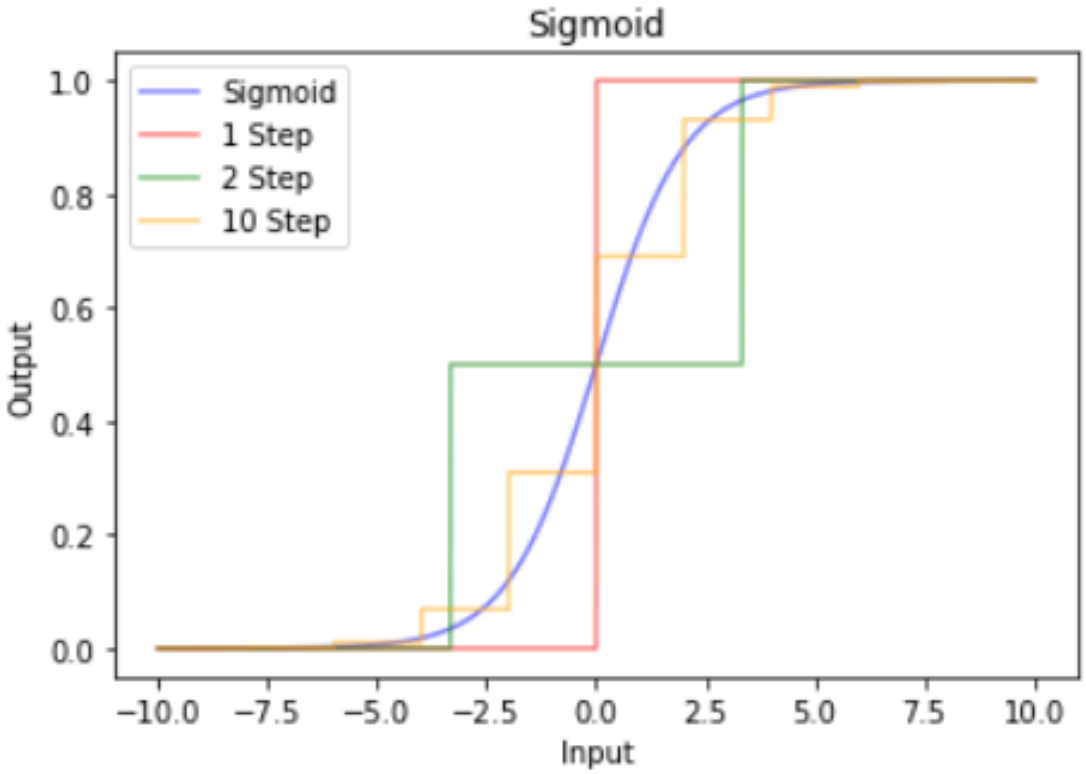
Step Approximation



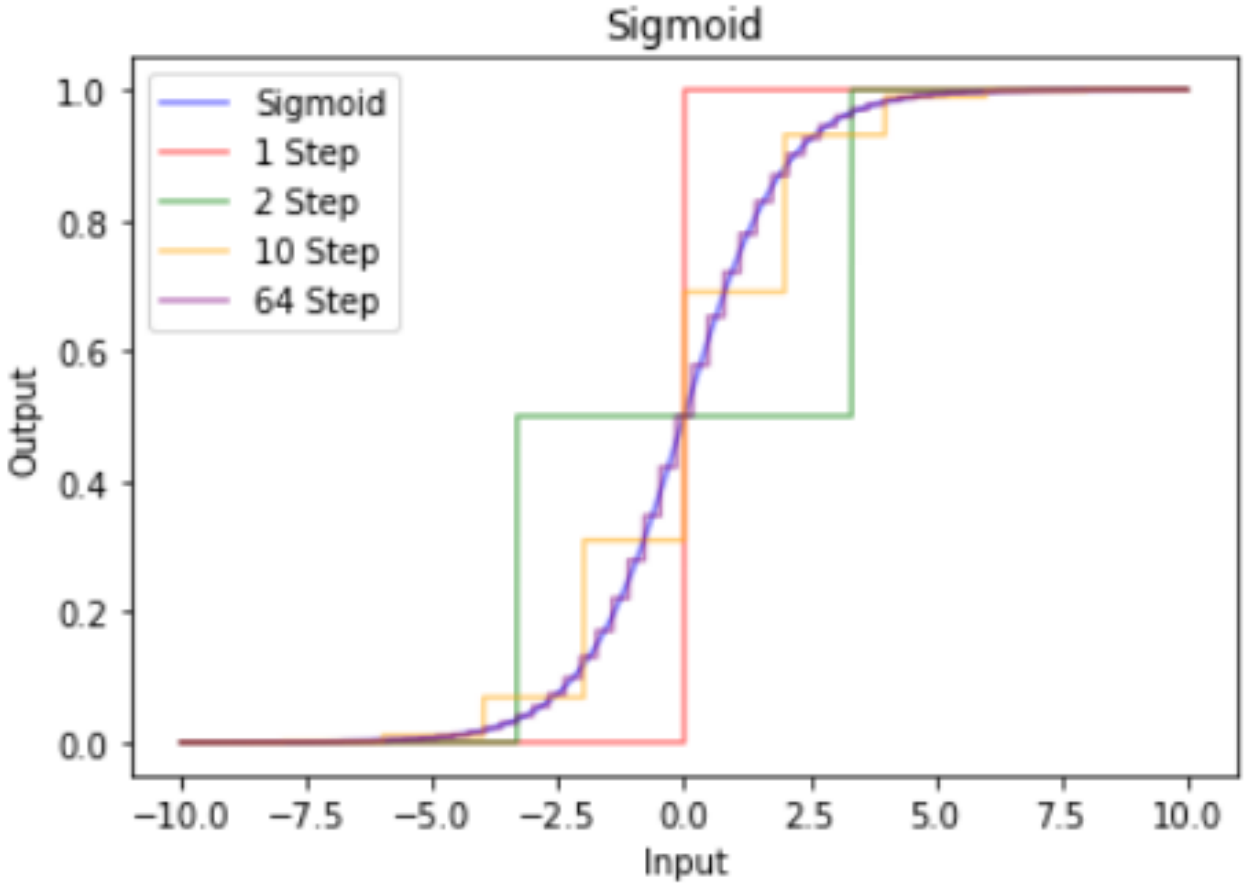
Multi-Step Approximation



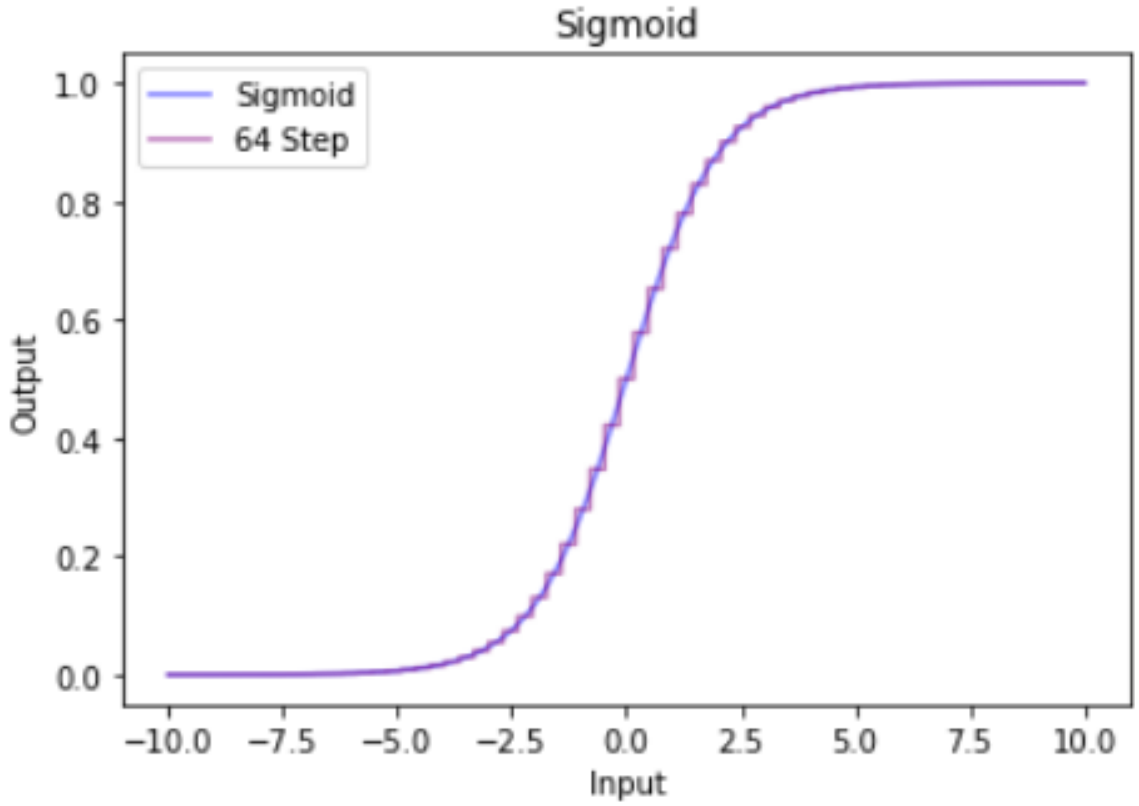
Many-Step Approximation

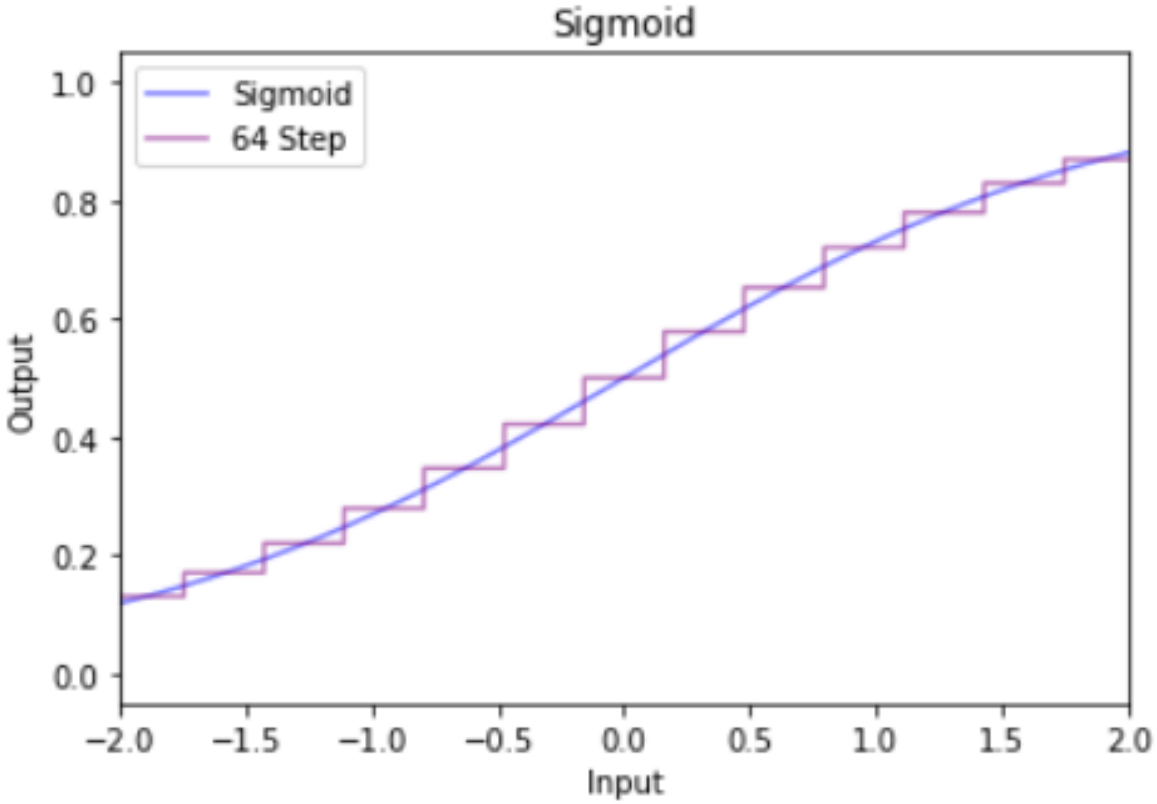


Many-Step Approximation

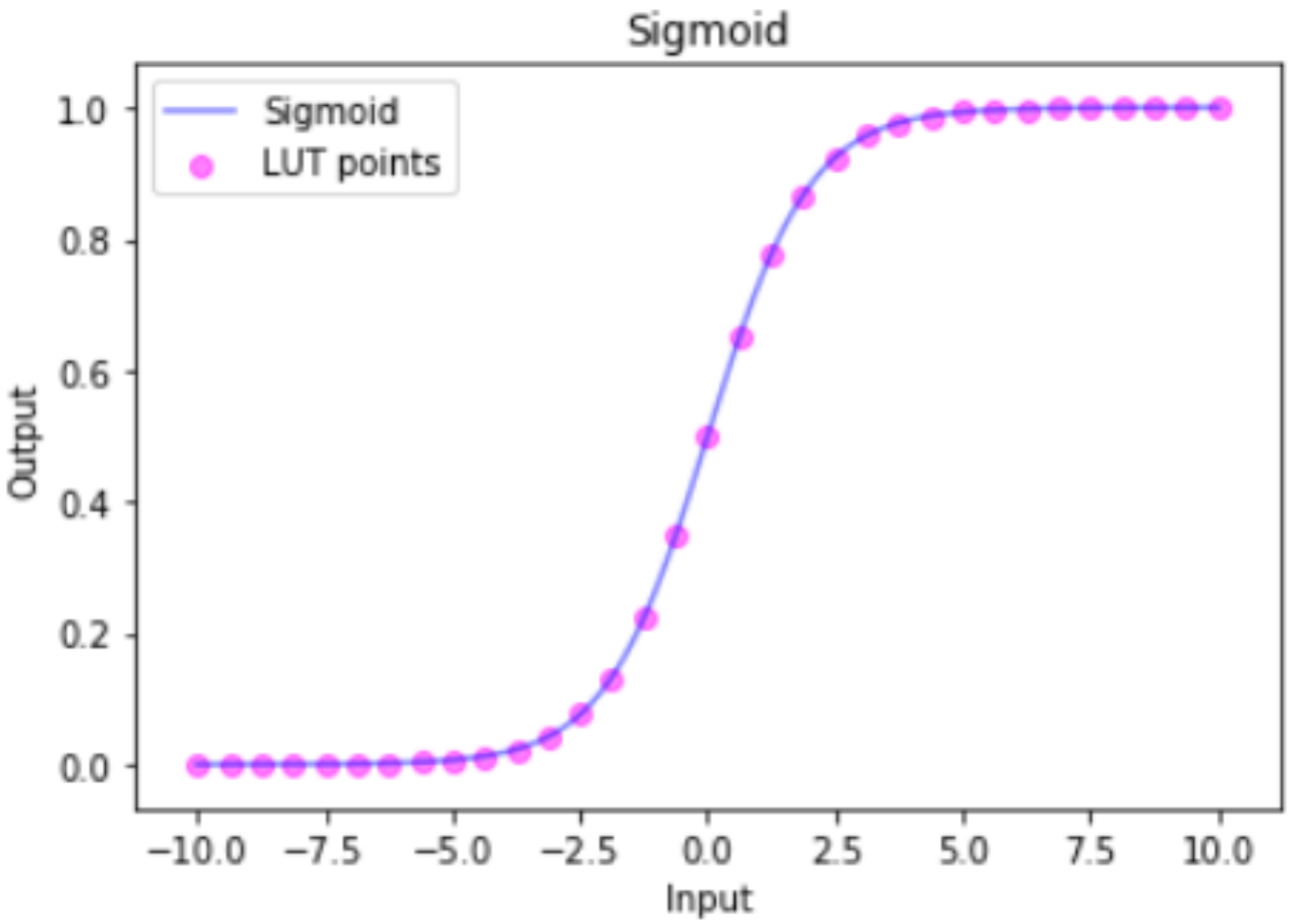


Approximated Sigmoid

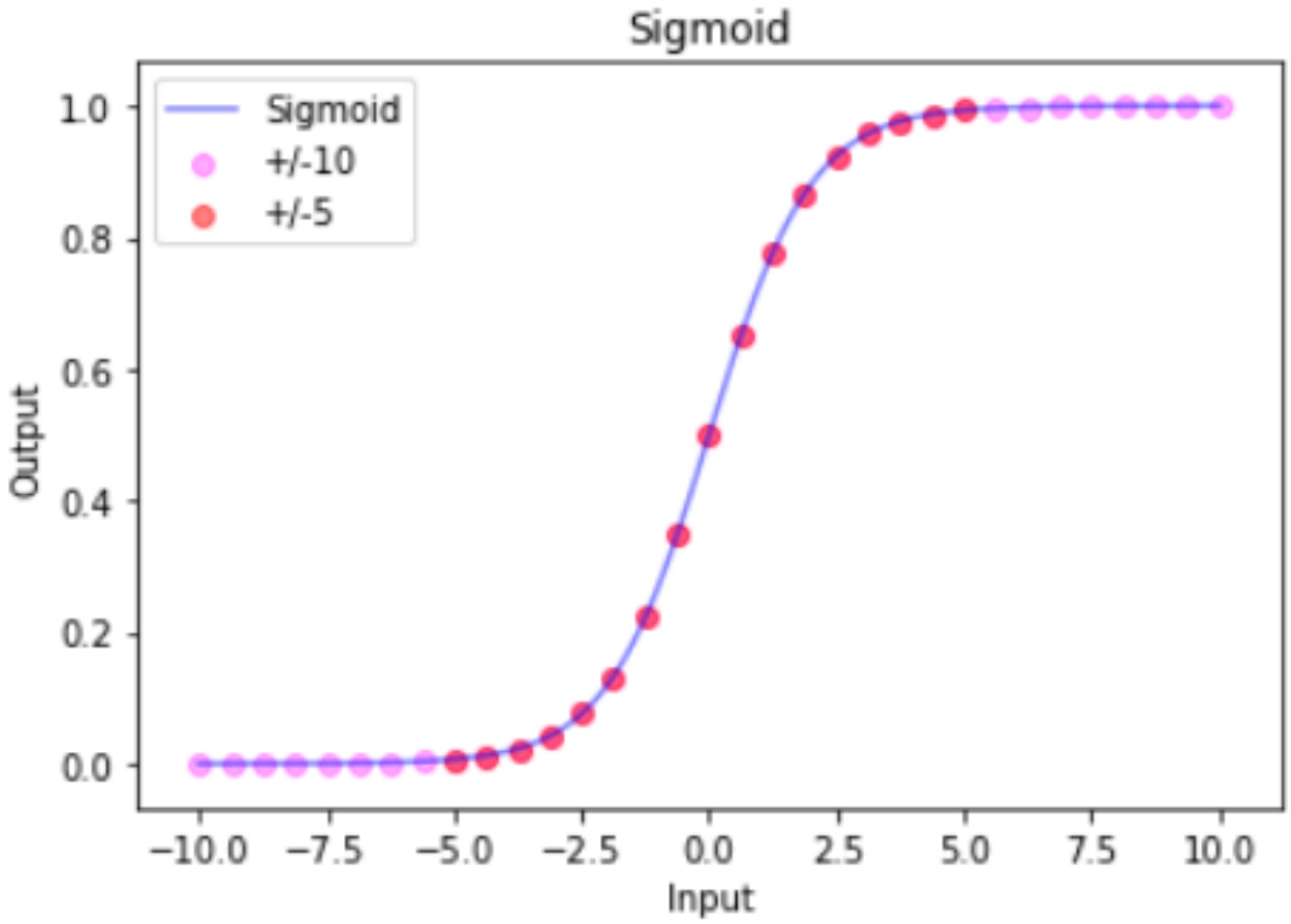




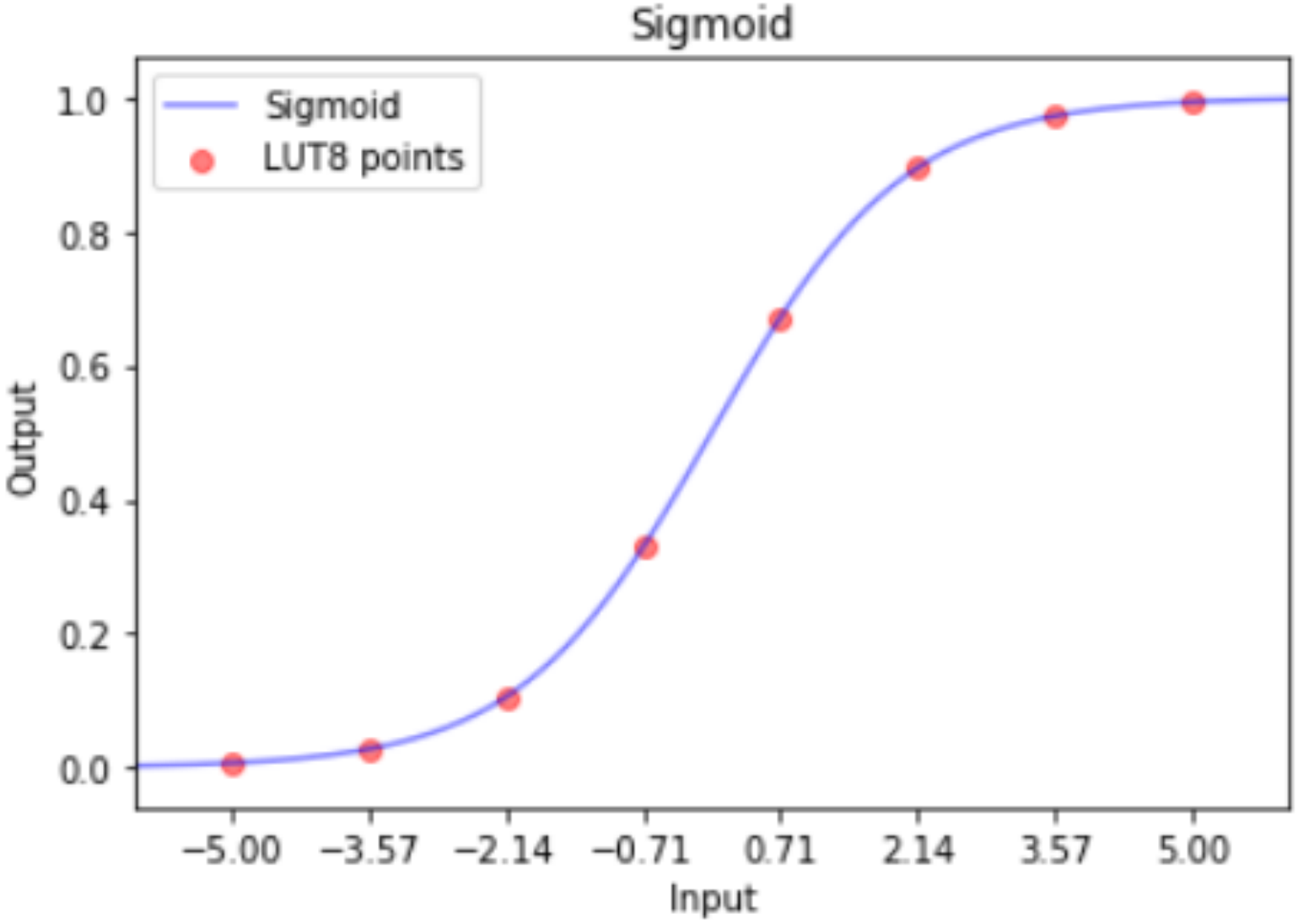
Pre-Calculate Sigmoid points



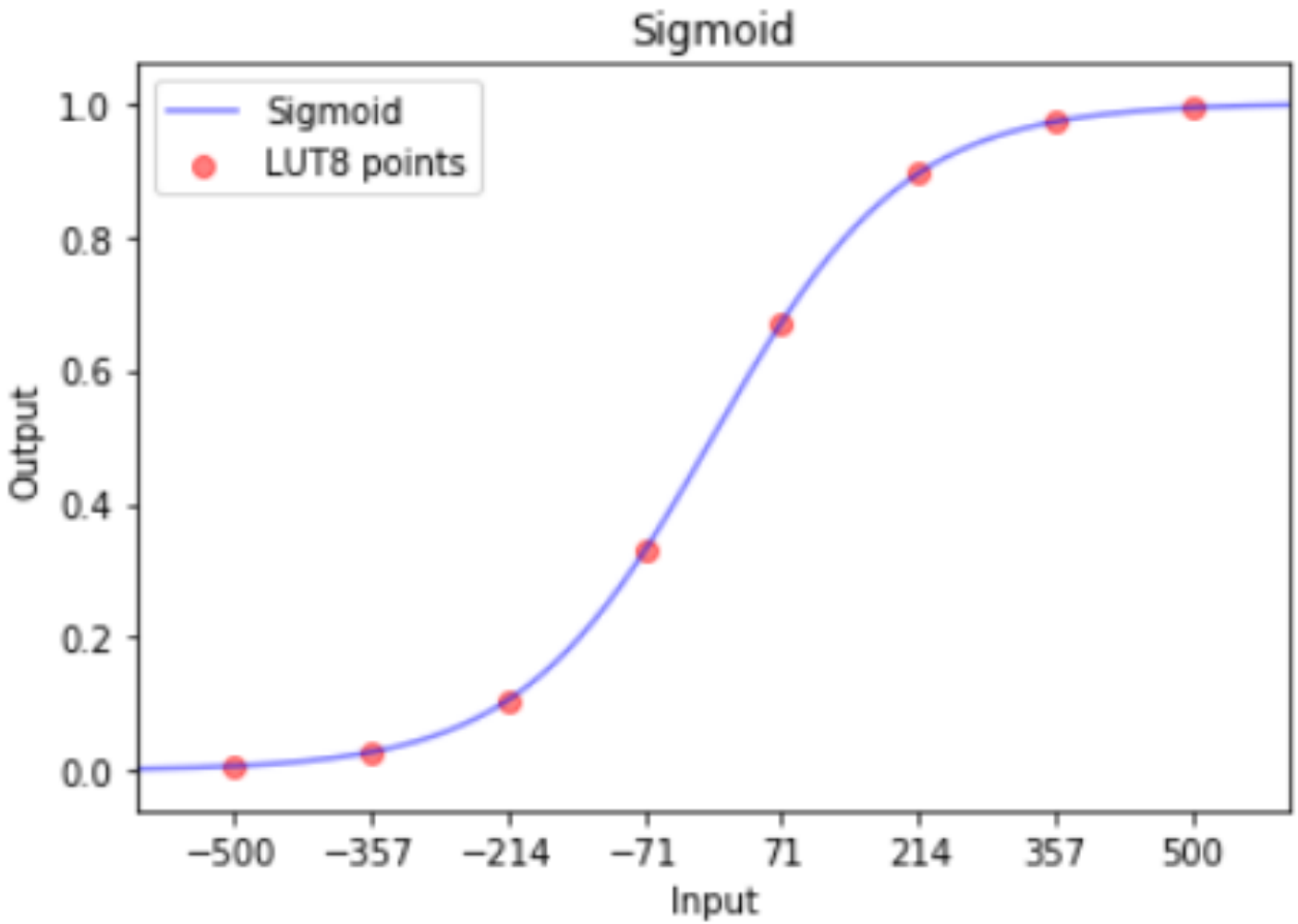
Optimization: Remove redundant values



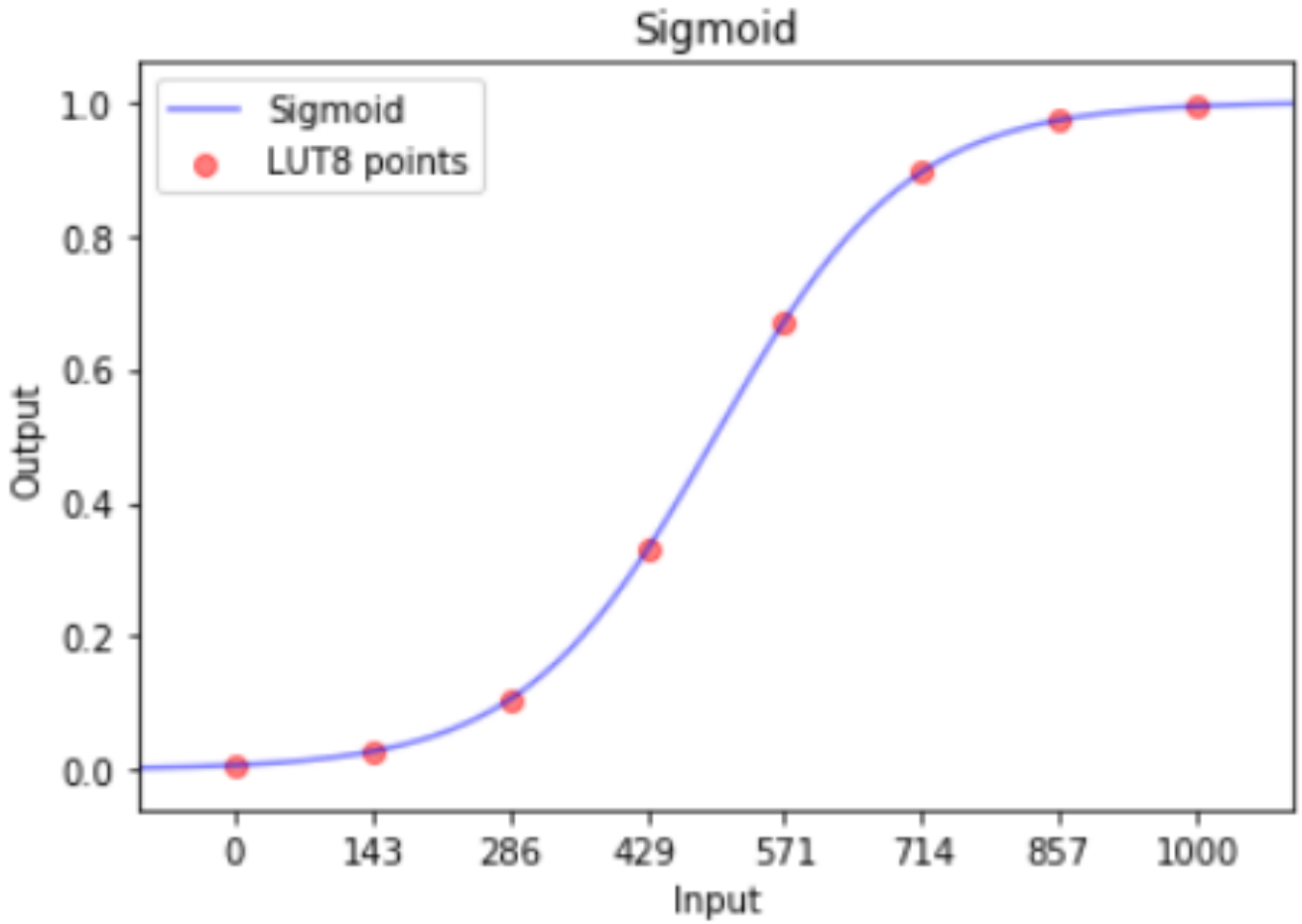
How do we find the nearest point?



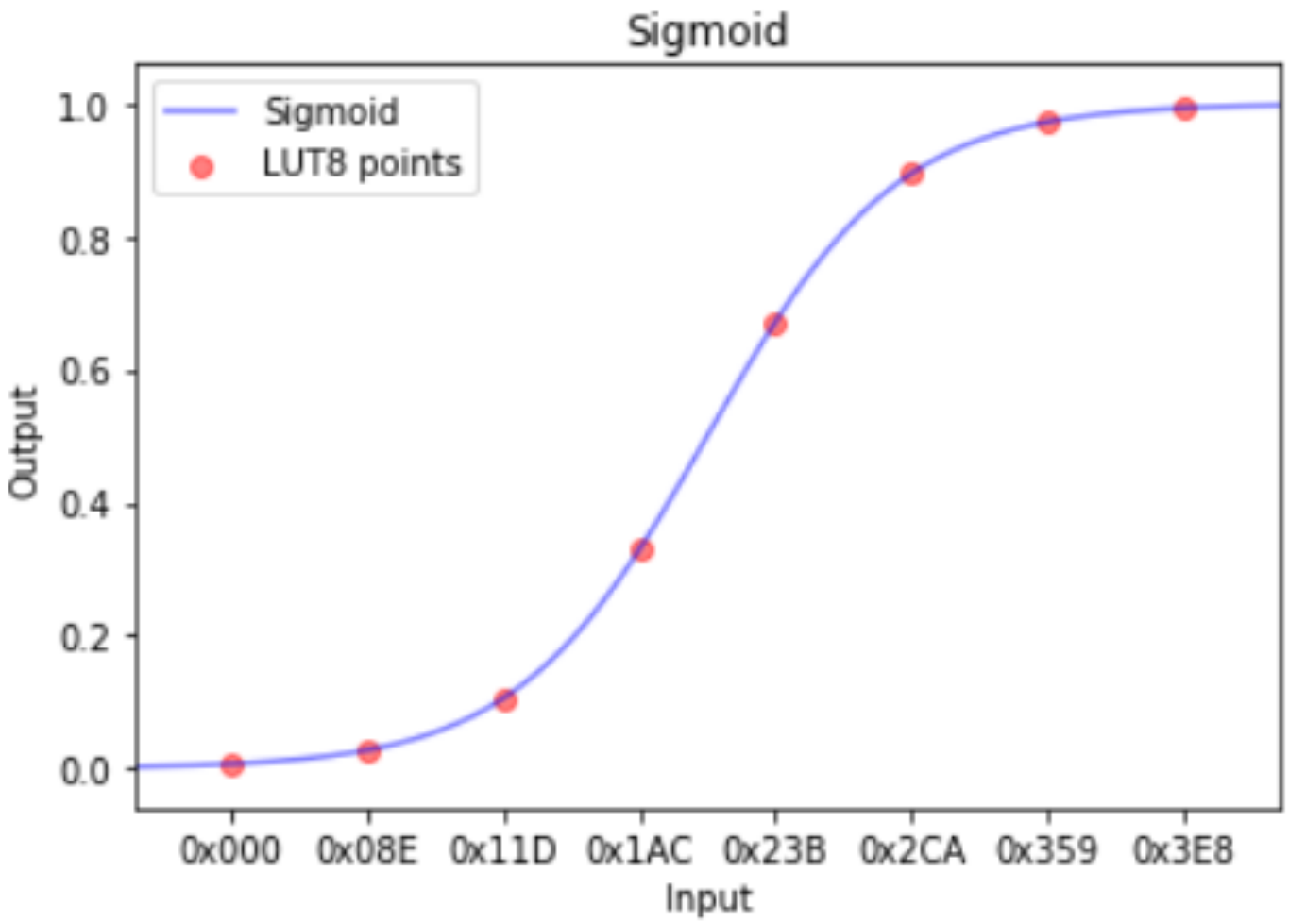
What about now? (X-axis scaled)



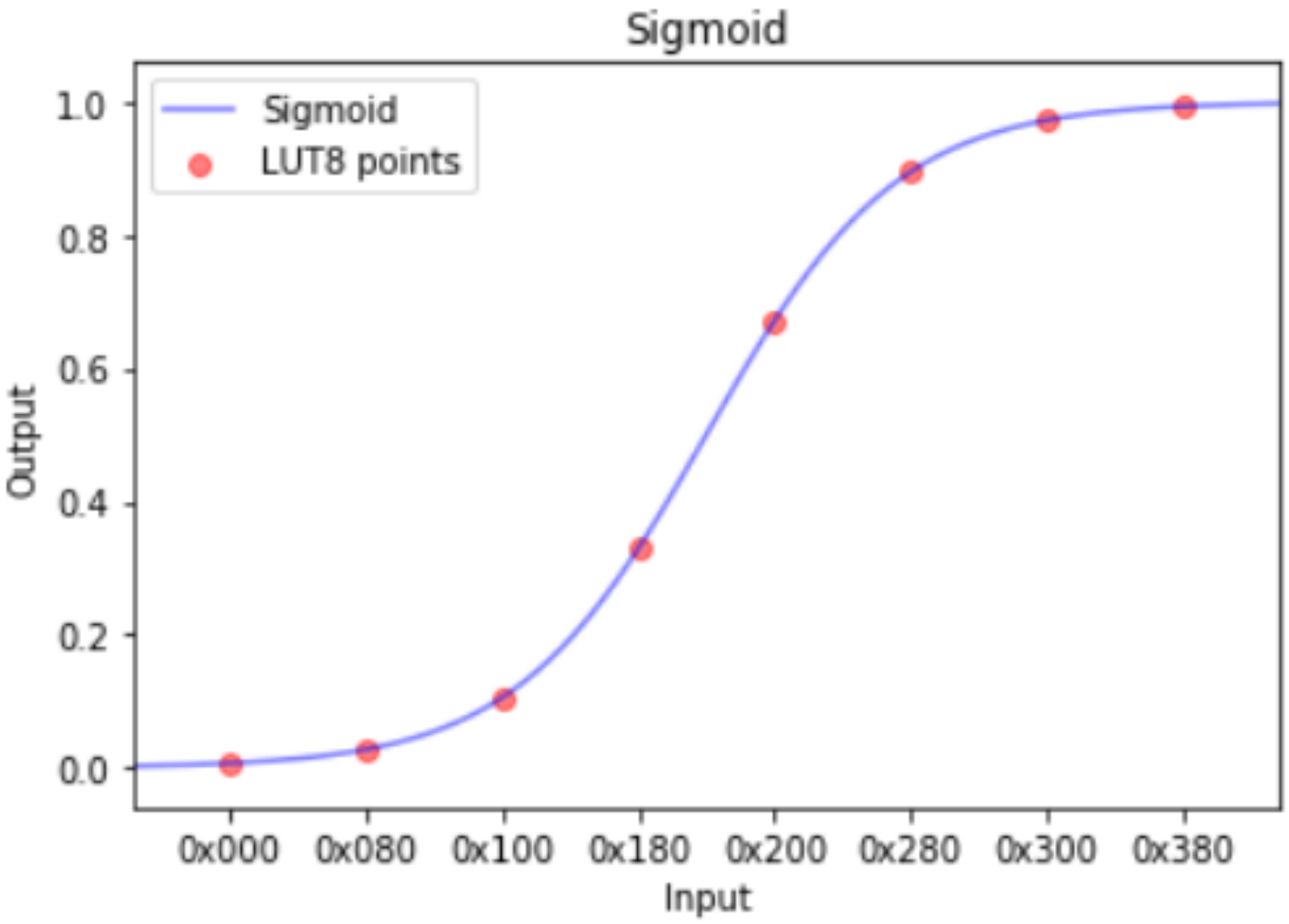
What about now? (X-axis shifted)



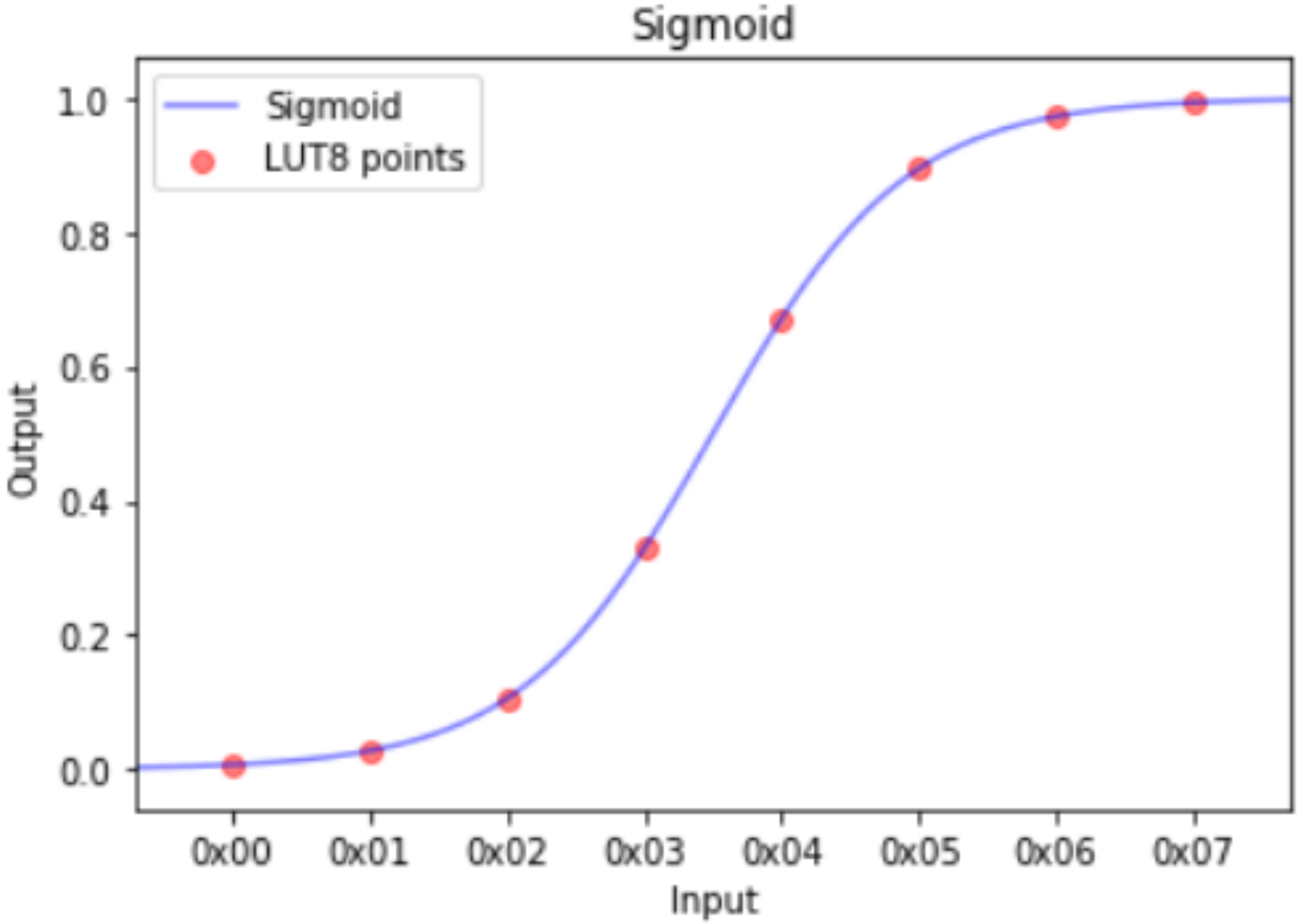
What about now? (X-axis radix)



What about now? (X-axis masked)



What about now? (X-axis bitshifted)



How do we generate an index?

Index	X	Y
0	-5.000	0.007
1	-3.571	0.027
2	-2.143	0.105
3	-0.714	0.329
4	0.714	0.671
5	2.143	0.895
6	3.571	0.973
7	5.000	0.993

How do we generate an index?

Index	X	Y
0	-5.000	0.007
1	-3.571	0.027
2	-2.143	0.105
3	-0.714	0.329
4	0.714	0.671
5	2.143	0.895
6	3.571	0.973
7	5.000	0.993

```
def gen_idx(x):
    return int((x + offset)/scale * arr_len)
```

```
scale2 = arr_len / scale
def gen_idx2(x):
    return int((x+offset) * scale2)
```

What about gen_idx2(-10)? Gen_idx2(+10)?

Index	X	Y
0	-5.000	0.007
1	-3.571	0.027
2	-2.143	0.105
3	-0.714	0.329
4	0.714	0.671
5	2.143	0.895
6	3.571	0.973
7	5.000	0.993

```
scale2 = arr_len / scale
def gen_idx2(x):
    return int((x+offset) * scale2)
```

What about gen_idx2(-10)? Gen_idx2(+10)?

Index	X	Y
0	-5.000	0.007
1	-3.571	0.027
2	-2.143	0.105
3	-0.714	0.329
4	0.714	0.671
5	2.143	0.895
6	3.571	0.973
7	5.000	0.993

```
scale2 = arr_len / scale
def gen_idx2(x):
    return int((x+offset) * scale2)
```

```
def gen_idx3(x):
    if (x <= -5): return 0
    elif (x >= +5): return 7
    else: return int((x+offset) * scale2)
```

FP Addition is expensive.

Index	X	Y
0	-5.000	0.007
1	-3.571	0.027
2	-2.143	0.105
3	-0.714	0.329
4	0.714	0.671
5	2.143	0.895
6	3.571	0.973
7	5.000	0.993

```
def gen_idx3(x):
    if (x <= -5): return 0
    elif (x >= +5): return 7
    else: return int((x+offset) * scale2)
```

FP Addition is expensive.

Index	X	Y
0	-5.000	0.007
1	-3.571	0.027
2	-2.143	0.105
3	-0.714	0.329
4	0.714	0.671
5	2.143	0.895
6	3.571	0.973
7	5.000	0.993

```python
def gen_idx3(x):
    if (x <= -5): return 0
    elif (x >= +5): return 7
    else: return int((x+offset) * scale2)
```

FP Addition is expensive.

Index	X	Y
0	-5.000	0.007
1	-3.571	0.027
2	-2.143	0.105
3	-0.714	0.329
4	0.714	0.671
5	2.143	0.895
6	3.571	0.973
7	5.000	0.993

```
def gen_idx3(x):
    if (x <= -5): return 0
    elif (x >= +5): return 7
    else: return int((x+offset) * scale2)
```

```
scale3 = int(scale2 * offset)
def gen_idx4(x):
    if (x <= -5): return 0
    elif (x >= +5): return 7
    else: return int(x*scale2) + scale3
```

```
def gen_idx5(x):
    idx = int(x*scale2) + scale3
    if (idx < 0): return 0
    if (idx > 7): return 7
    else: return idx
```

PX: Current Plan

- Shift to using tanh (not sigmoid)
- You get to build a LUT for tanh
- Still waiting to see if that will be faster. ☺

Float to Hex In Python

```python
import struct
def float_to_hex(f):
    return hex(struct.unpack('<I', struct.pack('<f', f))[0])

flts = [-16, -2.1, -1.2, 0, 1.2, 5.0, 32.999, 42.0]
flts_hex = list(map(float_to_hex, flts))
print (list(zip(flts, flts_hex)))
```

```
[(-16, '0xc1800000'), (-2.1, '0xc0066666'), (-1.2, '0xbf9999
9a'), (0, '0x0'), (1.2, '0x3f99999a'), (5.0, '0x40a00000'),
(32.999, '0x4203fefa'), (42.0, '0x42280000')]
```

19: Lookup Table Approximation

Engr 315: Hardware / Software Codesign

Andrew Lukefahr
Indiana University